Chapter 1: The OSI and TCP/IP Models

A network model divides communication into layers, where each layer provides services to the layer above it. The OSI reference model has seven layers: Physical, Data Link, Network, Transport, Session, Presentation and Application, from bottom to top.

The Physical layer sends raw bits over a medium. The Data Link layer sends frames between neighbouring devices using MAC addresses. The Network layer routes packets between networks using IP addresses. The Transport layer provides end-to-end delivery using ports, with TCP and UDP. The Session, Presentation and Application layers handle sessions, data formatting and encryption, and user-facing protocols such as HTTP and SMTP.

The TCP/IP model used on the Internet is simpler and has four layers: Link, Internet, Transport and Application. Encapsulation means that each layer adds its own header to the data as it moves down the stack, and the receiver removes the headers as the data moves up.

Chapter 2: IP Addressing and Subnetting

An IPv4 address is 32 bits long and is written as four decimal numbers separated by dots, for example 192.168.1.10. IPv4 provides about 4.3 billion addresses, which are nearly exhausted. IPv6 uses 128-bit addresses written in hexadecimal and solves the shortage.

An IP address has a network part and a host part. The subnet mask, or the CIDR prefix length, shows how many bits belong to the network. For example, 192.168.1.0/24 means the first 24 bits identify the network, which leaves 8 bits for hosts: 256 addresses in total, of which 254 are usable because the first address is the network address and the last is the broadcast address.

Private address ranges are not routed on the public Internet: 10.0.0.0/8, 172.16.0.0/12 and 192.168.0.0/16. Network Address Translation (NAT) lets many devices with private addresses share one public IP address. DHCP automatically assigns IP addresses to devices when they join a network.

Chapter 3: Transport Layer: TCP and UDP

The transport layer delivers data between applications on different hosts, and uses port numbers to identify the application. Port numbers range from 0 to 65535; for example HTTP uses port 80, HTTPS uses 443 and SSH uses 22.

TCP (Transmission Control Protocol) is connection-oriented and reliable. It sets up a connection with a three-way handshake (SYN, SYN-ACK, ACK), numbers every byte, retransmits lost segments, delivers data in order, and uses flow control and congestion control. TCP is used for web pages, email and file transfer.

UDP (User Datagram Protocol) is connectionless and unreliable: it sends datagrams without a handshake, without acknowledgements and without ordering. It has very low overhead, so it is used for DNS queries, video streaming, online games and voice calls, where speed matters more than perfect delivery.

Chapter 4: DNS and HTTP

The Domain Name System (DNS) translates human-readable domain names such as example.com into IP addresses. A DNS lookup goes from the local resolver to a root server, then to the top-level domain server (such as .com), and then to the authoritative name server of the domain. Resolvers cache answers for a time called the TTL to speed up later requests. Common record types are A (IPv4 address), AAAA (IPv6 address), CNAME (alias) and MX (mail server).

HTTP (HyperText Transfer Protocol) is the application-layer protocol of the web, based on a request-response model. The common methods are GET (read a resource), POST (send data to create something), PUT (replace a resource) and DELETE (remove a resource). HTTP is stateless, so each request is independent, and cookies are used to keep session information.

Status codes tell the result of a request: 200 OK means success, 301 means the resource moved permanently, 404 means not found, and 500 means an internal server error. In general, 2xx codes indicate success, 3xx redirection, 4xx client errors and 5xx server errors. HTTPS is HTTP carried over TLS, which encrypts the data.

Chapter 5: Switching and Routing

A switch works at the Data Link layer and connects devices inside one local area network (LAN). It learns which MAC address is reachable through which port and forwards each frame only to the correct port, which reduces collisions. A hub, in contrast, repeats every signal to all ports.

A router works at the Network layer and connects different networks. It uses a routing table to decide the next hop for each packet according to the destination IP address. Routes can be static, configured by hand, or dynamic, learned through routing protocols.

Routing protocols are divided into interior and exterior protocols. RIP is a distance-vector protocol that counts hops and allows at most 15 hops. OSPF is a link-state protocol that builds a map of the network and runs Dijkstra's algorithm to find the shortest paths. BGP is the exterior protocol that exchanges routes between autonomous systems and holds the global Internet together.

Chapter 6: Network Security

Network security protects data and systems from unauthorised access and attacks. The CIA triad describes its goals: Confidentiality (only authorised people can read data), Integrity (data is not altered) and Availability (systems are accessible when needed).

A firewall filters traffic according to rules based on IP addresses, ports and protocols. Transport Layer Security (TLS) provides encryption, authentication and integrity for data in transit and is the foundation of HTTPS. A Virtual Private Network (VPN) creates an encrypted tunnel across a public network, such as the Internet.

Common attacks include phishing, where users are tricked into giving up credentials; a denial-of-service (DoS) attack, which floods a server with traffic so that real users cannot reach it; a man-in-the-middle attack, where an attacker secretly intercepts communication; and ARP spoofing on local networks. Defences include strong passwords with multi-factor authentication, regular updates, encryption and network monitoring.